GRADE 9 MATHEMATICS
CONTINUOUS ASSESSMENT TEST 2

TERM 1 | TIME: 1 HOUR 15 MINUTES | TOTAL: 50 MARKS

Name: _____ Adm. No.: _____

School: _____ Date: _____

INSTRUCTIONS

- Answer all questions.
- Show all necessary working in calculations.
- A calculator may be used where appropriate.
- The marks for each question are shown in brackets.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Evaluate: $-18 + 7 \times 3$. (2 marks)

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2. Find the cube of 5. (2 marks)

_____

3. Simplify: $3^4 \times 3^2 \div 3^3$. (2 marks)

_____

4. Express 0.00072 in standard form. (2 marks)

_____

5. A worker completes a job in 12 days. How many days would 3 workers take at the same rate? (2 marks)

_____

6. Given A = [[2, 3], [1, 4]] and B = [[1, 0], [2, 5]], find A + B. (2 marks)

_____

7. Find the gradient of the line passing through (2, 3) and (6, 11). (2 marks)

_____

8. Solve the inequality: $2x + 5 < 13$. (2 marks)

_____

9. Find the area of a triangle with base 12 cm and height 7 cm. (2 marks)

_____

10. Convert 3.5 hours into minutes. (2 marks)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Indices and standard form (8 marks)

(a) Simplify, leaving your answer as a power: $2^5 \times 2^3 \div 2^4$. (3 marks)

(b) Evaluate: 27^(2/3). (2 marks)

(c) Calculate $(3.6 \times 10^5) \div (1.2 \times 10^2)$, giving your answer in standard form. (3 marks)

12. Proportion and rates of work (7 marks)

(a) Four taps fill a tank in 6 hours. How long would 3 identical taps take to fill the same tank? (4 marks)

(b) Eight workers complete a task in 15 days. How many days would 12 workers take, working at the same rate? (3 marks)

13. Matrices and straight-line graphs (8 marks)

(a) Given P = [[4, 1], [2, 3]] and Q = [[1, 2], [0, 5]], find 2P − Q. (4 marks)

(b) Find the equation of a straight line with gradient 3 and y-intercept −2. (2 marks)

(c) State the gradient and y-intercept of y = −2x + 5. (2 marks)

14. Measurement and data (7 marks)

(a) A rectangular floor is 8 m long and 5.5 m wide. Calculate its area. (2 marks)

(b) A journey of 180 km takes 2.5 hours. Calculate the average speed in km/h. (2 marks)

(c) The marks scored by five learners are 4, 7, 8, 6 and 10. Find the mean mark. (3 marks)

MARKING SCHEME – GRADE 9 MATHEMATICS CAT 2

Total: 50 marks

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. $-18 + 7 \times 3 = -18 + 21 = 3$. (2)

2. $5^3 = 125$. (2)

3. 3^(4+2−3) = $3^3 = 27$. (2)

4. 7.2×10^{-4}. (2)

5. $12 \div 3 = 4$ days. (2)

6. A + B = [[3, 3], [3, 9]]. (2)

7. Gradient = (11−3)/(6−2) = 8/4 = 2. (2)

8. $2x < 8$, hence $x < 4$. (2)

9. Area = ½ × 12 × 7 = 42 cm^2. (2)

10. 3.5 × 60 = 210 minutes. (2)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Indices and standard form

(a) 2^(5+3−4) = 2^4. (3: correct law 1, exponent 1, answer 1)

(b) 27^(2/3) = $(\sqrt[3]{27})^2 = 3^2 = 9$. (2)

(c) (3.6/1.2) × 10^(5−2) = 3×10^3. (3)

12. Proportion and rates of work

(a) Inverse proportion: $4 \times 6 = 3 \times t$; t = 24/3 = 8 hours. (4)

(b) $8 \times 15 = 12 \times d$; d = 120/12 = 10 days. (3)

13. Matrices and straight-line graphs

(a) 2P = [[8, 2], [4, 6]]; 2P − Q = [[7, 0], [4, 1]]. Award 2 marks for 2P and 2 for subtraction/result. (4)

(b) $y = 3x - 2$. (2)

(c) Gradient = −2; y-intercept = 5. (1 each)

14. Measurement and data

(a) Area = 8 × 5.5 = 44 m^2. (2)

(b) Speed = distance/time = 180/2.5 = 72 km/h. (2)

(c) Mean = (4+7+8+6+10)/5 = 35/5 = 7. (3: total 1, divide by 5 1, answer 1)

Teacher’s note: This is original teacher-prepared assessment material. Check that the topics assessed have been covered before administering. Accept equivalent correct methods and award method marks where shown.